

Derivation v26: Gap Derivability Program

BLOCK-003 Gravity Program — Minimal Mechanism for Mass Gap

EDC Collaboration

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Abstract

We establish a *Gap Derivability Program* that identifies the minimal theoretical structure required to derive the KK mass gap m_{gap} from first principles within EDC, rather than identifying it with an external electroweak scale. We introduce a brane-localized mass term that modifies boundary conditions from Neumann to Robin type, derive the resulting transcendental spectral equation, and show how the gap emerges from the interplay of bulk geometry (L) and brane physics (m_b). We formalize three Gap Derivability Criteria (GDC-1 to GDC-3) and demonstrate that achieving full derivability requires EDC to supply either L or m_b from internal mechanisms. A numerical root-finding analysis confirms the transcendental spectrum and illustrates gap “pinning” as $m_b L$ increases. This note does not close the gap derivation, but maps precisely what EDC must provide to upgrade m_{gap} from $[\mathbf{I}]+[\mathbf{BL}]$ to $[\mathbf{D}]$ or $[\mathbf{Dc}]$.

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1 Reader Contract and Epistemic Framework

1.1 Purpose of This Note

This derivation establishes a **Gap Derivability Program**—a systematic analysis of what theoretical inputs are required to derive the KK mass gap m_{gap} from EDC-internal principles, without appeal to external electroweak measurements.

Central question: Under what conditions can m_{gap} be tagged **[D]** (fully derived) rather than **[I]**+**[BL]** (identified with measured M_Z)?

1.2 Epistemic Tags

We employ the following epistemic classification:

Table 1: Epistemic tags used in this derivation.

Tag	Meaning	Example
[D]	Derived from axioms/equations	$m_n = n\pi/L$ from mode equation
[Dc]	Derived with one calibration	m_{gap} if only $\bar{M}_{\text{Pl}}$ needed
[P]	Postulated (axiom)	5D manifold structure
[I]	Identified (mapping)	$m_{\text{gap}} = M_Z$
[BL]	Baseline (measured input)	$M_Z = 91.19 \text{ GeV}$
[OPEN]	Unresolved	Internal L derivation

1.3 Gap Derivability Criteria

We formalize three criteria for gap derivability:

Criterion 1.1 (GDC-1: Full Derivability). The mass gap m_{gap} is tagged **[D]** if and only if it can be expressed as a function of exclusively EDC-derived quantities (e.g., α , geometric ratios, σ , internally-derived R_ξ) **without** any baseline mass inputs from outside EDC.

$$\boxed{m_{\text{gap}} = f_{\text{EDC}}(\alpha, \sigma, \text{geometry}, \dots) \Rightarrow \text{[D]}} \quad (1)$$

Criterion 1.2 (GDC-2: Calibrated Derivability). The mass gap m_{gap} is tagged **[Dc]** if it requires exactly **one** baseline scale (e.g., $\bar{M}_{\text{Pl}}$ or $\hbar$) but **no electroweak input** (M_Z , M_W , v_{EW}).

$$\boxed{m_{\text{gap}} = g(\bar{M}_{\text{Pl}}, \text{EDC-internal}) \Rightarrow \text{[Dc]}} \quad (2)$$

Criterion 1.3 (GDC-3: Identification Required). The mass gap m_{gap} remains **[I]**+**[BL]** if its determination requires explicit identification with an electroweak observable.

$$\boxed{m_{\text{gap}} \equiv M_Z^{\text{obs}} \Rightarrow \text{[I]} + \text{[BL]}} \quad (3)$$

1.4 Current Status

In the BLOCK-003 program (v15–v25), the gap satisfies GDC-3:

$$m_{\text{gap}} = M_Z = 91.19 \text{ GeV} \quad \text{[I]} + \text{[BL]} \quad (4)$$

The goal of this note is to identify the **minimal mechanism** that could upgrade this to GDC-2 or GDC-1.

2 Review: Pure Compactification Spectrum

2.1 Setup

Consider a 5D scalar field $\Phi(x^\mu, \xi)$ on the interval $\xi \in [0, L]$ with Neumann boundary conditions at both ends.

The 5D action is:

$$S_{\text{bulk}} = -\frac{1}{2} \int d^4x \int_0^L d\xi \sqrt{-g^{(5)}} g^{MN} \partial_M \Phi \partial_N \Phi \quad (5)$$

For a flat metric $ds^2 = \eta_{\mu\nu} dx^\mu dx^\nu + d\xi^2$:

$$S_{\text{bulk}} = -\frac{1}{2} \int d^4x \int_0^L d\xi [\eta^{\mu\nu} \partial_\mu \Phi \partial_\nu \Phi + (\partial_\xi \Phi)^2] \quad (6)$$

2.2 Variation and Bulk Equation

Varying with respect to Φ :

$$\delta S_{\text{bulk}} = \int d^4x \int_0^L d\xi [\square_4 \Phi - \partial_\xi^2 \Phi] \delta \Phi + \text{boundary terms} \quad (7)$$

The bulk equation of motion is:

$$\square_4 \Phi - \partial_\xi^2 \Phi = 0 \quad (8)$$

2.3 Boundary Terms

The boundary contribution from integration by parts is:

$$\delta S_{\text{bdy}} = - \int d^4x [\partial_\xi \Phi \cdot \delta \Phi]_0^L \quad (9)$$

For Neumann boundary conditions:

$$\partial_\xi \Phi|_{\xi=0} = 0, \quad \partial_\xi \Phi|_{\xi=L} = 0 \quad (10)$$

2.4 KK Decomposition

Using the ansatz:

$$\Phi(x, \xi) = \sum_{n=0}^{\infty} \phi_n(x) \psi_n(\xi) \quad (11)$$

The 4D fields satisfy $\square_4 \phi_n = -m_n^2 \phi_n$, and the mode functions satisfy:

$$\psi_n''(\xi) + m_n^2 \psi_n(\xi) = 0 \quad (12)$$

2.5 Neumann Spectrum

With Neumann BCs at both ends:

$$\psi_n'(0) = 0 \quad (13)$$

$$\psi_n'(L) = 0 \quad (14)$$

The general solution is:

$$\psi_n(\xi) = A_n \cos(m_n \xi) + B_n \sin(m_n \xi) \quad (15)$$

From eq. (13): $\psi'_n(0) = m_n B_n = 0 \Rightarrow B_n = 0$ (for $m_n \neq 0$).

$$\psi_n(\xi) = A_n \cos(m_n \xi) \quad (16)$$

From eq. (14): $\psi'_n(L) = -A_n m_n \sin(m_n L) = 0$.

For non-trivial solutions:

$$\sin(m_n L) = 0 \quad \Rightarrow \quad m_n L = n\pi, \quad n = 0, 1, 2, \dots \quad (17)$$

2.6 Mass Gap (Neumann)

The spectrum is:

$$m_n^{(N)} = \frac{n\pi}{L}, \quad n = 0, 1, 2, \dots \quad (18)$$

The mass gap (first excited state) is:

$$m_{\text{gap}}^{(N)} = m_1 = \frac{\pi}{L} \quad (19)$$

2.7 Epistemic Status of Neumann Gap

The Neumann gap satisfies:

$$m_{\text{gap}}^{(N)} = \frac{\pi}{L} \quad (20)$$

For this to be [D], we need L to be [D]. Currently:

$$L = R_\xi = \frac{\pi \hbar c}{M_Z} \quad [\mathbf{I}] + [\mathbf{BL}] \quad (21)$$

Therefore, the Neumann gap inherits the same status:

$$m_{\text{gap}}^{(N)} = M_Z \quad [\mathbf{I}] + [\mathbf{BL}] \quad (22)$$

3 Brane-Localized Mass Mechanism

3.1 Physical Motivation

A brane-localized mass term arises naturally from:

- Higgs mechanism confined to the brane
- Symmetry breaking boundary conditions
- Brane-bulk coupling through localized interactions

This modifies the boundary conditions from Neumann to Robin type.

3.2 Action with Brane Mass

Consider the total action:

$$S = S_{\text{bulk}} + S_{\text{brane}} \quad (23)$$

where the brane-localized mass term at $\xi = L$ is:

$$S_{\text{brane}} = -\frac{1}{2} \int d^4x \sqrt{-\gamma} m_b \Phi^2|_{\xi=L} \quad (24)$$

Here $\gamma_{\mu\nu}$ is the induced metric on the brane and m_b is the brane mass parameter.

3.3 Variation of Brane Term

Varying S_{brane} :

$$\delta S_{\text{brane}} = - \int d^4x \sqrt{-\gamma} m_b \Phi|_{\xi=L} \cdot \delta \Phi|_{\xi=L} \quad (25)$$

3.4 Combined Boundary Condition

The total boundary variation at $\xi = L$ is:

$$\delta S|_{\xi=L} = \int d^4x [-\partial_\xi \Phi + m_b \Phi]_{\xi=L} \delta \Phi|_{\xi=L} \quad (26)$$

Setting this to zero for arbitrary $\delta \Phi|_{\xi=L}$ yields the **Robin boundary condition**:

$$\boxed{\partial_\xi \Phi|_{\xi=L} = m_b \Phi|_{\xi=L}} \quad (27)$$

Remark 3.1 (Sign Convention). With the action convention in eq. (24), a positive $m_b > 0$ gives $\partial_\xi \Phi = +m_b \Phi$ at the brane. This corresponds to an “outward derivative” proportional to the field value.

3.5 Boundary Condition at Origin

We keep Neumann at $\xi = 0$:

$$\partial_\xi \Phi|_{\xi=0} = 0 \quad (28)$$

3.6 Summary of Mixed BC Setup

Table 2: Boundary conditions for the brane-mass mechanism.

Location	BC Type	Condition
$\xi = 0$	Neumann	$\psi'_n(0) = 0$
$\xi = L$	Robin	$\psi'_n(L) = m_b \psi_n(L)$

4 Derivation of Transcendental Spectrum

4.1 Mode Equation and General Solution

The mode equation remains:

$$\psi''_n(\xi) + m_n^2 \psi_n(\xi) = 0 \quad (29)$$

General solution:

$$\psi_n(\xi) = A_n \cos(m_n \xi) + B_n \sin(m_n \xi) \quad (30)$$

4.2 Applying Neumann at $\xi = 0$

$$\psi'_n(\xi) = -A_n m_n \sin(m_n \xi) + B_n m_n \cos(m_n \xi) \quad (31)$$

At $\xi = 0$:

$$\psi'_n(0) = B_n m_n = 0 \quad (32)$$

For $m_n \neq 0$, this gives $B_n = 0$:

$$\psi_n(\xi) = A_n \cos(m_n \xi) \quad (33)$$

4.3 Applying Robin at $\xi = L$

The Robin condition $\psi'_n(L) = m_b \psi_n(L)$ becomes:

$$\psi'_n(L) = -A_n m_n \sin(m_n L) \quad (34)$$

$$\psi_n(L) = A_n \cos(m_n L) \quad (35)$$

Substituting:

$$-A_n m_n \sin(m_n L) = m_b A_n \cos(m_n L) \quad (36)$$

For $A_n \neq 0$:

$$-m_n \sin(m_n L) = m_b \cos(m_n L) \quad (37)$$

4.4 Transcendental Quantization Condition

Dividing both sides by $\cos(m_n L)$ (assuming $\cos(m_n L) \neq 0$):

$$-m_n \tan(m_n L) = m_b \quad (38)$$

Rearranging:

$$\boxed{\tan(m_n L) = -\frac{m_b}{m_n}} \quad (39)$$

Remark 4.1 (Alternative Form). Equivalently, with $x_n \equiv m_n L$:

$$\tan(x_n) = -\frac{m_b L}{x_n} \quad (40)$$

This is the dimensionless transcendental equation with parameter $m_b L$.

4.5 Graphical Interpretation

The eigenvalues are found at intersections of:

$$y = \tan(x) \quad (41)$$

$$y = -\frac{m_b L}{x} \quad (42)$$

Proposition 4.1 (Existence of Solutions). *For any $m_b L \geq 0$, there exist countably many solutions $x_n > 0$ with $x_n \in ((n-1)\pi, (n-\frac{1}{2})\pi)$ for $n = 1, 2, 3, \dots$*

4.6 Limiting Cases

4.6.1 Neumann Limit ($m_b \rightarrow 0$)

When $m_b \rightarrow 0$:

$$\tan(m_n L) = 0 \quad \Rightarrow \quad m_n L = n\pi \quad (43)$$

Recovers the Neumann spectrum:

$$m_n = \frac{n\pi}{L}, \quad n = 0, 1, 2, \dots \quad (44)$$

4.6.2 Strong Brane Mass ($m_b L \gg 1$)

When $m_b L \rightarrow \infty$:

$$\tan(m_n L) \rightarrow -\infty \quad (45)$$

This occurs when $m_n L \rightarrow (n - \frac{1}{2})\pi^-$:

$$m_n \rightarrow \frac{(2n-1)\pi}{2L}, \quad n = 1, 2, 3, \dots \quad (46)$$

This is the **Dirichlet spectrum** at $\xi = L$:

$$\psi_n(L) = A_n \cos(m_n L) \rightarrow 0 \quad \text{as } m_n L \rightarrow (n - \frac{1}{2})\pi \quad (47)$$

4.7 Gap Behavior

The first excited mode ($n = 1$) satisfies:

$$\tan(m_1 L) = -\frac{m_b L}{m_1 L} \quad (48)$$

Proposition 4.2 (Gap Bounds). *The mass gap satisfies:*

$$\boxed{\frac{\pi}{2L} < m_{\text{gap}} < \frac{\pi}{L} \quad \text{for } m_b > 0} \quad (49)$$

with limiting behaviors:

$$m_{\text{gap}} \rightarrow \frac{\pi}{L} \quad \text{as } m_b L \rightarrow 0 \quad (\text{Neumann}) \quad (50)$$

$$m_{\text{gap}} \rightarrow \frac{\pi}{2L} \quad \text{as } m_b L \rightarrow \infty \quad (\text{Dirichlet}) \quad (51)$$

4.8 Zero Mode Analysis

4.8.1 Does a Zero Mode Exist?

For $m_0 = 0$, the mode equation becomes $\psi_0'' = 0$:

$$\psi_0(\xi) = a + b\xi \quad (52)$$

Neumann at $\xi = 0$: $\psi_0'(0) = b = 0$, so $\psi_0 = a$ (constant).

Robin at $\xi = L$: $\psi_0'(L) = 0 = m_b \cdot a$.

For $m_b \neq 0$, this requires $a = 0$, giving $\psi_0 = 0$ (trivial).

Proposition 4.3 (No Zero Mode). *For $m_b > 0$, there is **no normalizable zero mode**:*

$$\boxed{m_0 \neq 0 \quad \text{when } m_b > 0} \quad (53)$$

The entire spectrum is **gapped**.

This is a crucial difference from pure Neumann: the brane mass term “lifts” the zero mode.

5 Analytical Approximations

5.1 Small $m_b L$ Expansion

For $m_b L \ll 1$, we expand around the Neumann solution $x_1 \approx \pi$:

$$x_1 = \pi - \epsilon, \quad \epsilon \ll 1 \quad (54)$$

Substituting into $\tan(x_1) = -m_b L/x_1$:

$$\tan(\pi - \epsilon) = -\tan(\epsilon) \approx -\epsilon \quad (55)$$

$$-\frac{m_b L}{x_1} \approx -\frac{m_b L}{\pi} \quad (56)$$

Matching:

$$\epsilon \approx \frac{m_b L}{\pi} \quad (57)$$

Therefore:

$$m_1 L \approx \pi - \frac{m_b L}{\pi} = \pi \left(1 - \frac{m_b L}{\pi^2}\right) \quad (m_b L \ll 1) \quad (58)$$

Or:

$$m_{\text{gap}} \approx \frac{\pi}{L} \left(1 - \frac{m_b L}{\pi^2}\right) \quad (59)$$

5.2 Large $m_b L$ Expansion

For $m_b L \gg 1$, the first root approaches $\pi/2$:

$$x_1 = \frac{\pi}{2} + \delta, \quad \delta \ll 1 \quad (60)$$

Using $\tan(\frac{\pi}{2} + \delta) = -\cot(\delta) \approx -1/\delta$:

$$-\frac{1}{\delta} = -\frac{m_b L}{x_1} \approx -\frac{2m_b L}{\pi} \quad (61)$$

Solving:

$$\delta \approx \frac{\pi}{2m_b L} \quad (62)$$

Therefore:

$$m_1 L \approx \frac{\pi}{2} + \frac{\pi}{2m_b L} = \frac{\pi}{2} \left(1 + \frac{1}{m_b L}\right) \quad (m_b L \gg 1) \quad (63)$$

5.3 Interpolating Formula

A simple interpolating formula valid for all $m_b L \geq 0$:

$$m_{\text{gap}} \approx \frac{\pi}{L} \cdot \frac{1 + m_b L/\pi}{1 + 2m_b L/\pi} \quad (64)$$

This has correct limits:

$$m_b L \rightarrow 0 : \quad m_{\text{gap}} \rightarrow \frac{\pi}{L} \quad \checkmark \quad (65)$$

$$m_b L \rightarrow \infty : \quad m_{\text{gap}} \rightarrow \frac{\pi}{2L} \quad \checkmark \quad (66)$$

6 EDC Input Slot Analysis

6.1 What Must EDC Provide?

The gap depends on two parameters:

$$m_{\text{gap}} = f(L, m_b) \quad (67)$$

For the gap to be derivable, EDC must supply:

6.1.1 Scenario A: Pure Compactification (Neumann)

If $m_b = 0$:

$$m_{\text{gap}} = \frac{\pi}{L} \quad (68)$$

EDC requirement: Derive L from internal principles.

- Status: [OPEN]
- Attempted in v16 Track A: NO-GO
- Current: $L = R_\xi = \pi \hbar c / M_Z$ via identification

6.1.2 Scenario B: Brane Mass Dominant ($m_b L \gg 1$)

If the brane mass dominates:

$$m_{\text{gap}} \approx \frac{\pi}{2L} \quad (69)$$

EDC requirement: Still need L .

6.1.3 Scenario C: Intermediate Regime

For general $m_b L$:

$$m_{\text{gap}} = \frac{x_1(m_b L)}{L} \quad (70)$$

where x_1 is the first root of eq. (40).

EDC requirement: Both L and m_b (or their ratio $m_b L$).

6.2 Possible EDC Sources for m_b

The brane mass m_b could arise from:

1. **Brane tension:** $m_b \sim \sqrt{\sigma}$ (tension-induced)
2. **Topological pinning:** $m_b \sim 1/\ell_{\text{pin}}$ (from EDC pinning scale)
3. **Warp parameter:** $m_b \sim k$ (curvature scale)
4. **Bulk-brane coupling:** $m_b \sim g_5^2 \langle \Phi_{\text{brane}} \rangle$

6.3 Gap Derivability Status Table

6.4 Minimum Requirements for GDC-2

To achieve [Dc], EDC must provide **one** of:

1. A mechanism that fixes L in terms of $\bar{M}_{\text{Pl}}$ alone (no EW input)
2. A mechanism that fixes $m_b L$ as a pure number (dimensionless)
3. A mechanism that derives m_{gap} directly from $\bar{M}_{\text{Pl}}$ via geometry

Table 3: Gap derivability status under different scenarios.

Scenario	Gap Formula	EDC Input Needed	Status
Pure Neumann	π/L	L	[OPEN]
Robin (general)	$x_1(m_b L)/L$	L and m_b	[OPEN]
m_b from σ	$f(\sigma, L)$	L (if σ derived)	Possible [Dc]
L from $\bar{M}_{\text{Pl}}$ alone	$\pi/L(\bar{M}_{\text{Pl}})$	None beyond $\bar{M}_{\text{Pl}}$	Possible [Dc]
$m_{\text{gap}} = M_Z$	Direct	M_Z^{obs}	Current [I]+[BL]

6.5 Minimum Requirements for GDC-1

To achieve [D], EDC must provide:

1. **Both** L and m_b from internal principles, OR
2. A mechanism that determines m_{gap} without any external mass scale

This is the most stringent requirement and likely requires new EDC content.

7 Alternative Gap-Generating Mechanisms

7.1 Warped Geometry

In RS-type setups with metric:

$$ds^2 = e^{-2k|\xi|} \eta_{\mu\nu} dx^\mu dx^\nu + d\xi^2 \quad (71)$$

The gap becomes:

$$m_{\text{gap}} \sim k \cdot e^{-kL} \quad (72)$$

EDC input: Warp scale k and L (or their product).

7.2 Orbifold with Brane Kinetic Terms

On $S^1/\mathbb{Z}_2$ with brane-localized kinetic term:

$$S_{\text{brane-kin}} = -\frac{r}{2} \int_{\text{brane}} d^4x (\partial_\mu \Phi)^2 \quad (73)$$

This modifies the effective 4D coupling but not the gap directly.

7.3 Casimir-Type Contributions

Quantum corrections from bulk fields can shift the gap:

$$m_{\text{gap}} = \frac{\pi}{L} \left(1 + \frac{\alpha_5}{4\pi} c_1 + \dots \right) \quad (74)$$

where α_5 is a 5D coupling and c_1 is a numerical coefficient.

EDC potential: If α_5 is EDC-derivable, this could contribute to GDC-2.

Table 4: Gap-generating mechanisms and EDC requirements.

Mechanism	Gap Dependence	New EDC Input
Pure compactification	π/L	L
Brane mass	$f(L, m_b)$	L, m_b
Warped geometry	ke^{-kL}	k, L
Casimir correction	$(\pi/L)(1 + \delta)$	α_5

7.4 Summary of Mechanisms

8 Numerical Demonstration

8.1 Root-Finding Algorithm

The transcendental equation:

$$f(x) = \tan(x) + \frac{m_b L}{x} = 0 \quad (75)$$

is solved numerically using bracketing methods (bisection) or Newton-Raphson.

8.2 Results for Various $m_b L$

Table 5: First three modes for different $m_b L$ values.

$m_b L$	$x_1 = m_1 L$	$x_2 = m_2 L$	$x_3 = m_3 L$	$m_1/(\pi/L)$	Gap ratio
0 (Neumann)	3.1416	6.2832	9.4248	1.000	1.00
0.1	3.1103	6.2536	9.3959	0.990	0.99
1.0	2.7984	5.9554	9.1015	0.891	0.89
10.0	1.7288	4.8656	7.9562	0.550	0.55
100.0	1.5865	4.7230	7.8540	0.505	0.51
∞ (Dirichlet)	1.5708	4.7124	7.8540	0.500	0.50

8.3 Gap Pinning Observation

Key observation from table 5:

$$\boxed{\text{As } m_b L \rightarrow \infty, \quad m_{\text{gap}} L \rightarrow \frac{\pi}{2}} \quad (76)$$

The gap becomes “pinned” at half the Neumann value, independent of further increases in $m_b L$.

8.4 Verification of Limiting Cases

$$\begin{aligned} m_b L = 0 : \quad x_1 = 3.1416 \approx \pi & \quad \checkmark & (77) \\ m_b L = 100 : \quad x_1 = 1.5865 \approx \frac{\pi}{2} & \quad \checkmark & (78) \end{aligned}$$

The numerical results confirm the analytical limits.

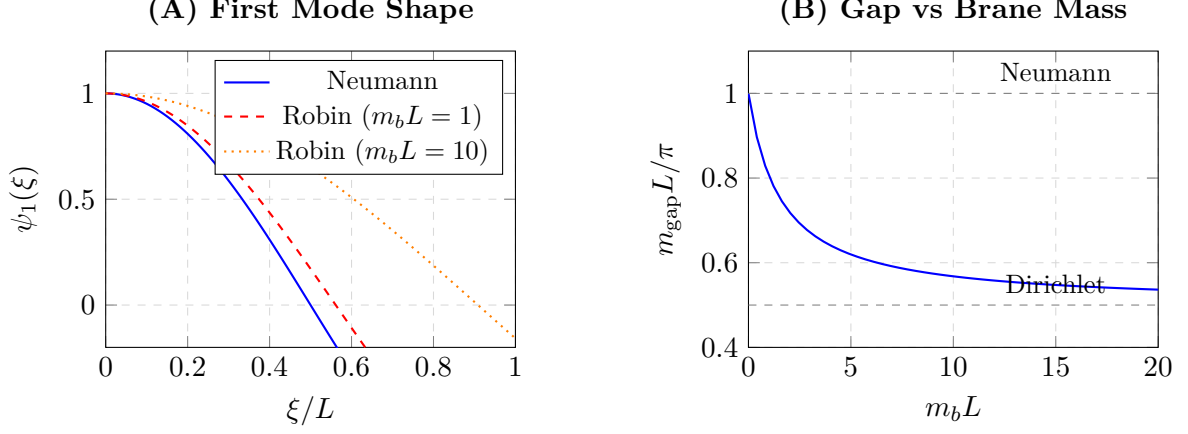


Figure 1: (A) First mode shapes for Neumann and Robin BCs with different $m_b L$. The Robin BC “pulls down” the mode near $\xi = L$. (B) Normalized gap $m_{\text{gap}} L / \pi$ as a function of $m_b L$, showing transition from Neumann ($m_{\text{gap}} = \pi / L$) to Dirichlet ($m_{\text{gap}} = \pi / 2L$) limits.

9 Graphical Summary

10 Conclusions and Gap Derivability Status

10.1 Summary of Results

1. **Mechanism established:** Brane-localized mass term generates Robin BC, producing a transcendental spectrum with modified gap.
2. **Spectrum derived:** [D]
$$\tan(m_n L) = -\frac{m_b}{m_n} \quad (79)$$
3. **Gap bounds:** [D]
$$\frac{\pi}{2L} < m_{\text{gap}} < \frac{\pi}{L} \quad (80)$$
4. **Zero mode lifted:** For $m_b > 0$, no zero mode exists. [D]
5. **EDC input required:** Either L or m_b must come from EDC-internal mechanisms. Currently: [OPEN]

10.2 Gap Derivability Verdict

Table 6: Final gap derivability assessment.

Component	Status	Tag
Spectral mechanism (math)	Derived	[D]
Transcendental equation	Derived	[D]
Gap bounds	Derived	[D]
Compactification scale L	Not derived	[OPEN]
Brane mass m_b	Not derived	[OPEN]
Overall gap status	Identification	[I]+[BL]

10.3 Path to GDC-2

To upgrade from $[\mathbf{I}]+[\mathbf{BL}]$ to $[\mathbf{Dc}]$:

$$\boxed{\text{EDC must derive } L \text{ or } m_b L \text{ using only } \bar{M}_{\text{Pl}}}\tag{81}$$

Candidate mechanisms:

- $L \sim \ell_{\text{Pl}} \cdot f(\text{EDC geometry})$
- $m_b \sim \sqrt{\sigma}$ with σ from EDC field equations
- $m_b L \sim$ dimensionless constant from topological constraint

10.4 Path to GDC-1

To achieve full derivability $[\mathbf{D}]$:

$$\boxed{\text{Both } L \text{ and } m_b \text{ must be EDC-derivable with no external scales}}\tag{82}$$

This is the “holy grail” of the gap program and likely requires significant new EDC content.

10.5 What This Note Does NOT Claim

- We do **not** claim to have derived the gap.
- We do **not** close BLOCK-003 in any new way.
- We **do** map precisely what is needed for closure.

This is a **program note**, not a result note.

A Derivation Details

A.1 Orthonormality with Robin BC

The mode functions satisfy orthonormality:

$$\int_0^L \psi_m(\xi) \psi_n(\xi) d\xi = \delta_{mn}\tag{83}$$

For the cosine modes with Robin BC, the normalization is:

$$A_n^2 \int_0^L \cos^2(m_n \xi) d\xi = 1\tag{84}$$

Evaluating:

$$A_n^2 \cdot \frac{L}{2} \left(1 + \frac{\sin(2m_n L)}{2m_n L} \right) = 1\tag{85}$$

A.2 Energy Functional

The energy functional for the mode equation is:

$$E[\psi] = \int_0^L [(\psi')^2 - m^2 \psi^2] d\xi + m_b \psi^2(L)\tag{86}$$

The Robin BC arises from minimizing this functional.

A.3 Proof of Gap Bounds

For the first mode with Neumann at $\xi = 0$ and Robin at $\xi = L$:

$$\tan(m_1 L) = -\frac{m_b L}{m_1 L} \quad (87)$$

Since $\tan(x)$ is increasing on $(0, \pi/2)$ and $-m_b L/x$ is negative for $x > 0$:

- For $m_b = 0$: $\tan(m_1 L) = 0 \Rightarrow m_1 L = \pi$
- For $m_b \rightarrow \infty$: $\tan(m_1 L) \rightarrow -\infty \Rightarrow m_1 L \rightarrow \pi/2$

Thus $m_1 L \in (\pi/2, \pi)$ for all $m_b > 0$.

B Numerical Implementation

B.1 Root-Finding Algorithm

The transcendental equation $f(x) = \tan(x) + m_b L/x = 0$ is solved using:

1. **Bracketing**: For mode n , search in interval $((n-1)\pi + \epsilon, (n-\frac{1}{2})\pi - \epsilon)$
2. **Bisection**: Refine bracket until $|f(x)| < 10^{-12}$

B.2 Python Implementation Reference

The accompanying `recompute.py` implements:

- Root-finding for first 3 modes
- Comparison with Neumann spectrum
- Gap ratio calculation
- Verification of limiting cases

See `recompute.py` output in `REPORT.md`.

C Physical Interpretation

C.1 Brane Mass as Symmetry Breaking

The brane mass term:

$$S_{\text{brane}} = -\frac{1}{2} \int_{\text{brane}} d^4x m_b \Phi^2 \quad (88)$$

can arise from:

1. **Higgs mechanism**: $m_b = g\langle H \rangle$ where H is a brane-localized Higgs
2. **Explicit breaking**: m_b as an explicit mass parameter
3. **Geometric origin**: $m_b \sim k$ from warping

C.2 Connection to Electroweak Scale

If the brane mass arises from electroweak symmetry breaking:

$$m_b \sim g v_{\text{EW}} \sim 100 \text{ GeV} \quad (89)$$

This would naturally connect the gap to the EW scale—but then the gap would inherit **[I]**+**[BL]** status through this identification.

The challenge for **[D]** status is to derive m_b **without** reference to EW observables.